MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

1/ TYPE DOD-015

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tunable frequency 8,800 to 9,500 MHz, pulsed type, integral magnet, 400 watts minimum peak power output.

ABSOLUTE RATINGS:

Parameter:	Ef	tk	tpc	rrv	Du	σ'	ib	epy	pi	TE	Alt
Unit:	V	sec	μs	kv/μs	---	---	a	kv	kw	°C	ft
Maximum:	5.5	---	1.0	12	0.002	1.5	1.3	2.4	3.1	125	200,000
Minimum:	4.5	30	0.3	---	---	---	---	2.0	---	---	---
	(Note 1)	(Note 1)		(Note 2)							(Note 3)

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Output coupling:	Coaxial TNC
Mounting position:	*Any*	*Weight:*	*7 ounces (max)*
Mounting support:	See figure 1 and note 4	Cooling:	Convection (see note 4)
Input connections:	Wire leads (see figure 1)		

TEST CONDITIONS:

Parameter:	Ef	Du	tpc	rrv	ib	Ib	epy	VSWR	F
Unit:	V	---	μs	kv/μs	a	mAdc	kv	---	MHz
Tolerance:	---	---	±0.2	min	---	---	---	max	---
	5.0	0.0012	0.8	10	1.1	1.32	2.1	1.1:1	F1,F2,F3

Frequency		
F	MHz	±
1	8,800	0.1%
2	9,150	0.1%
3	9,500	0.1%

GENERAL:

(A) First article sample approval - Required (see note 23)

1/ Replaces Motorola part number 95-21486B01 and Varian part number BLM-111.

(A) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
Ⓐ	First article inspection						
1016	Moisture resistance	-		---	---	---	---
1031	Variable-frequency vibration (operating)	19,20		ΔF	---	±2.0	MHz
1042	Shock, specified pulse	5	Test condition I	ΔF	---	±2.0	MHz
1047	High - and low - temperature performance (operating)	4,6,7	TA = -55° to +90°C	---	---	---	---
4027	Temperature coefficient	4,6, 7,8		ΔF/°C	---	0.05	MHz/°C
4028	Barometric pressure, reduced	3,4		---	---	---	---
4304	Pulse characteristics (duty cycle drift)	4,9		ΔF	---	6.0	MHz
---	Constant acceleration	10	75 G; t = 5 minutes	ΔF	---	±2.0	MHz
---	Thermal equilibrium	4,6,11		ΔF	---	3.0	MHz
---	Rotation	12		ΔF	---	±2.0	MHz
	Quality conformance inspection, part 1	13					
4003	Pressurizing	14	Method A; 30 psia	---	---	---	---
4289	Heater current (nonoperating)	-	Ef = 5.0 V; tk = 30 (min)	If	0.65	0.75	A
4303	Heater warmup time	-	Ef = 5.0 V	tk	---	30	sec
4250	Power output	15		po	400	---	w
4306	Pulse voltage	-		epy	2.0	2.4	kv
4223	Mechanical tuning range	15		F	8,800	9,500	MHz
4308	RF bandwidth	-		BW	---	3.0/tpc	MHz
4315	Stability	16	prr = 150 and 2,000	MP	---	0.25	%
	Quality conformance inspection, part 2						
4310	*Frequency pulling figure*	15,17	σ = 1.5:1 *(min)*	ΔF	---	15	MHz
4311	Frequency pushing figure	18		ΔF	---	5.3	MHz
1031	Variable frequency vibration (operating)	19,20	F = F1; t = 5 minutes; plane B	ΔF	---	±2.0	MHz
	Quality conformance inspection, part 3						
---	Life test	21	Group D; F = F2	t	250	---	hrs
---	Life-test end points:						
4250	Power output	15		po	300	---	w
4308	RF bandwidth	-		BW	---	3.0/tpc	MHz
4315	Stability	16	prr = 150 and 2,000	MP	---	0.50	%
---	Frequency change	22		ΔF	---	8.0	MHz

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NOTES:

1. The pulse voltage shall be applied only when the heater-cathode warmup time has elapsed.

2. The rate of rise voltage (rrv) shall be measured in accordance with method 4304, except that the steepest tangent to the leading edge of the voltage pulse shall be measured above the 80 percent amplitude point. Any capacitance used in the viewing circuit shall not exceed 3 pF.

3. The TUT shall operate satisfactorily when its coaxial (output) end is exposed at altitudes up to 200,000 feet above sea level without degradation of performance. During this test, the input terminals shall be exposed at altitudes up to 15,000 feet.

4. The TUT shall be securely mounted to a rigid and flat aluminum mounting surface such that the maximum thermal conduction is obtained. The TUT shall be capable of being oriented in any direction when mounted in this manner.

5. The TUT shall be subjected to three shocks in each direction of the three mutually perpendicular planes for a total of 18 shocks. Following the 18 shocks, the TUT shall be capable of satisfying all requirements of this specification without damage or failure before, during, and after being subjected to the specified shock test.

6. The TUT shall be mounted on a heat sink during all temperature tests. The ambient temperature shall be defined as temperature of the heat sink. The air surrounding the TUT shall not be artificially circulated and shall be within ±10°C of the heat sink temperature.

7. The TUT shall meet all requirements of power output and spectrum measurements within the specified ambient temperature range.

8. Temperature coefficient shall be obtained by measuring the tube frequency after 10 minutes of operation following 45 minutes of stabilization at one ambient temperature, and then changing the ambient temperature 30°C with a similar temperature stabilization followed by 10 minutes of operation before the second frequency is measured. The temperature coefficient shall be within the limits specified regardless of where the 30°C change in ambient temperature occurs within the specified operating ambient temperature range.

9. The absolute value of the change in generated radio frequency shall not exceed the specified value as the prr is varied from 150 to 2,000, with peak pulse current held constant. The TUT frequency shall be stabilized at a prr of 150, 500, 1,000, and 2,000 in determining the duty cycle drift. All other conditions shall remain fixed.

10. The TUT shall be exposed to an acceleration of 75G for 5 minutes along each of the three mutually perpendicular axis. The frequency shift shall not exceed the specified limits and the spectrum shall exhibit no evidence of degradation of performance.

11. The TUT shall be allowed to cool at ambient temperature for a minimum of 1 hour. Following the cooling period, the filament will be preheated for a minimum of 30 seconds, then high voltage shall be applied. A frequency measurement shall be made 5 minutes after high voltage is applied. Frequency measurements shall be made periodically for 1 hour after the application of high voltage and the maximum frequency drift shall be less than the value specified.

12. The frequency shift shall not exceed the specified value when the TUT is rotated about its longitudinal axis at 300 revolutions/minute.

13. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

14. The seal formed by clamping the TUT mounting plate against a suitable magnetron test fixture, shall be hermetically tight with the air pressure specified herein applied to surround the entire input bushing below the mounting plate. A suitable "O" ring shall be used between the TUT mounting plate and test fixture.

15. The minimum specified peak power of the TUT shall be maintained over the entire frequency range with a VSWR of 1.1:1.

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NOTES: -Continued

16. Stability shall be measured in terms of the average number of missing output pulses expressed as a percentage of the number of input pulses applied during the period of observation. The missing pulses (MP) due to any causes, are considered to be missing if the rf energy is less than 70 percent of the normal energy level. The stability shall be measured when a VSWR of 1.5 minimum is intoduced into a load at a distance of 1 ±0.05 meter from the TUT flange and the phase is adjusted to produce maximum instability. The missing pulse test shall be counted during any three consecutive minute intervals of a 6-minute test period.

17. When varied throughout all phases, the frequency deviation (maximum frequency minus minimum frequency) shall not exceed the limits specified herein when a VSWR of 1.5:1 minimum is introduced into the load at a distance of 1 ±0.05 meter from the TUT flange.

18. The pushing figure shall be determined from the total frequency change when the average anode current is varied ±10 percent.

19. The TUT shall be fastened rigidly to the vibration platform and vibrated along each of its mutually perpendicular planes at a constant octave sweep from 5 to 2,000 to 5 Hz and shall require 25 minutes to transverse the range in each direction. The amplitude level shall be 0.4-inch double amplitude below 25 Hz and 15G above 25 Hz. The frequency shift shall not exceed the specified limits and there shall be no evidence of degradation of performance during the vibration cycle.

20. After the performance of the specified test, the TUT shall satisfy all requirements for rf bandwidth and pulse stability as specified in quality conformance inspection, part 1.

21. Life test shall be conducted in accordance with the following cycle. Life test hours shall be the total acceptable operating hours accumulated under condition B below.

Condition	Duration	Ef	Ib
	Minutes	V	mAdc
A	0.8	5.0	0
B	20.0	5.0	1.32
C	9.2	0	0

22. The frequency change during life shall be within the value specified.

(A)23. First article sample inspection shall conform to the requirements of appendix F of MIL-E-1 and shall consist of performing all tests specified on this tube specification sheet. Invitation for bids should provide that the preparing activity reserves the right to waive the requirements for first article samples as to those bidders offering a product which has been previous procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MU
Air Force - 80
DSA - ES

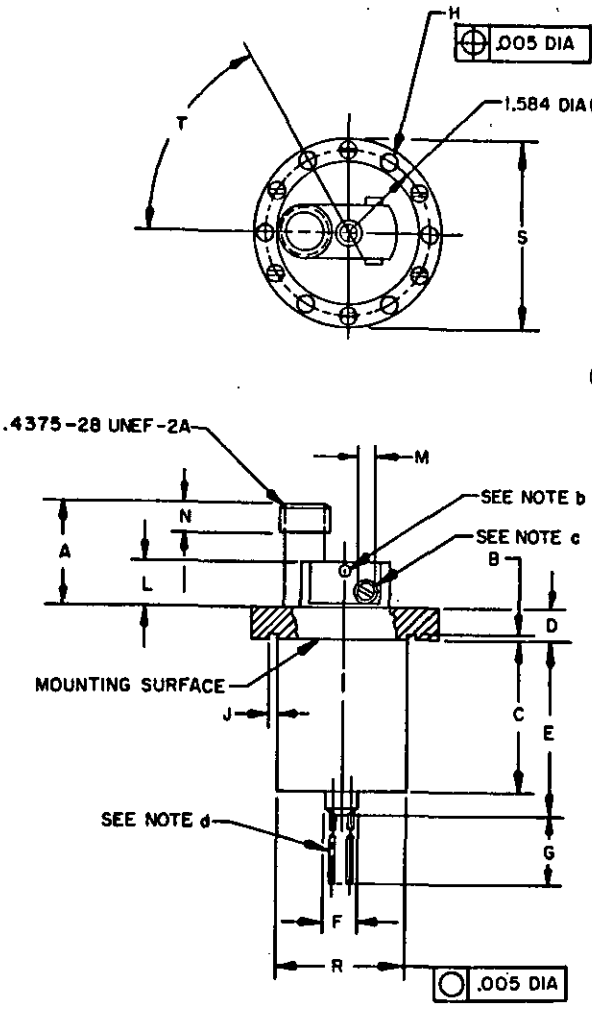
User activities:
Army - WC
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3106)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Ⓐ	First article inspection	
A	.820 (20.83)	.850(21.59)
B	.015 (.38)	.025 (.64)
C	1.33 (33.78)	1.41 (35.81)
T	58°	62°
	Quality conformance inspection, part 1	
D	.225 (5.72)	.245 (6.22)
E		1.575(40.01)
F		.900(22.86)
G	4.00 (101.60)	
H	.150 (3.81)	.154 (3.91)
J	.071 (1.80)	.081 (2.06)
	Quality conformance inspection, part 2	
L	.448 (11.38)	.478(12.14)
M	.115 (2.92)	.135 (3.43)
N	.250 (6.35)	
R	1.245 (31.62)	1.265(32.13)
S	1.798 (45.67)	1.818(46.81)

NOTES:
a. Pattern orientation for hole.
b. Tuner lock screw (2) required.
c. Tuning screw (2) required.
d. Input connections shall consist of No. 20 strand copper wire leads 4-inches (101.6 mm) long (minimum) that are teflon coated, or equal. A black wire shall be used for the cathode connection.

FIGURE 1. Outline drawing of electron tube type DOD-015.

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